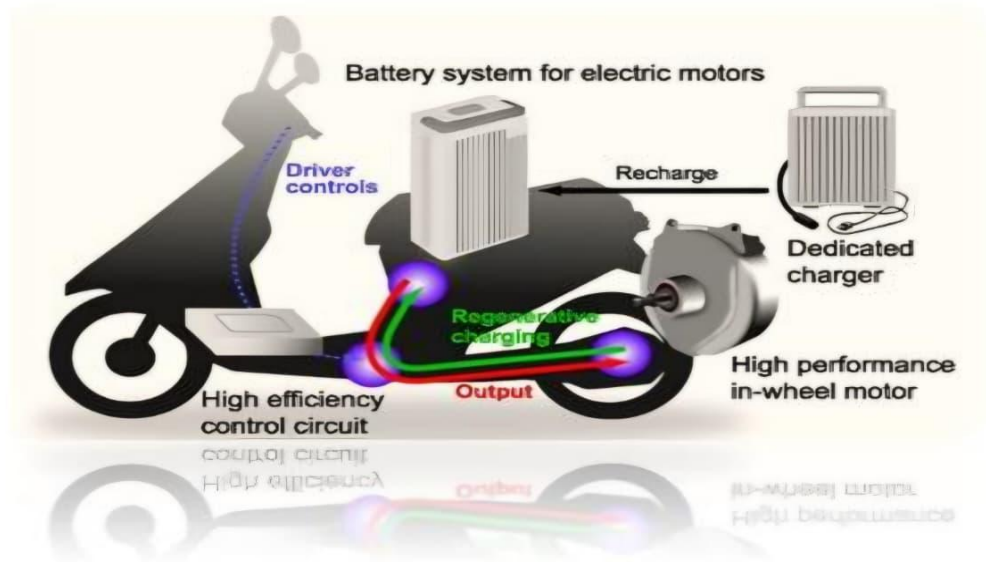


DESIGN & PROJECT PORTFOLIO

Mechanical Engineering



Guduru Sudeep

B.Tech – Mechanical Engineering

Key Skills

- CATIA V5
- AutoCAD
- ANSYS (FEA)

Email: sudeepchowdar27@gmail.com

Phone:7816057828

India

Introduction

I am Guduru Sudeep, B.Tech Mechanical Engineering graduate with a strong interest in Automotive Engineering and design. This portfolio presents a selection of my academic, design, and industrial experiences, highlighting my skills in mechanical design, manufacturing processes, and vehicle systems.

I have gained hands-on industrial exposure through an internship at Megha Engineering & Infrastructures Ltd. (MEIL), where I worked with conventional and CNC machining operations and participated in quality control and inspection activities using standard measuring instruments. This experience strengthened my understanding of manufacturing practices, tolerances, and engineering drawings.

In addition, my experience at KIA Motors service provided practical insight into automotive systems, vehicle maintenance, diagnostics, and real-world vehicle operation challenges, further reinforcing my interest in the automotive domain.

Academically, my work includes CATIA V5 design projects, and my final year major project on the design and fabrication of an electric vehicle, reflecting my inclination toward electric mobility and sustainable transportation.

I am motivated to pursue a Master's degree in Automotive Engineering, aiming to build expertise in vehicle design, electric powertrains, and advanced manufacturing, and to contribute to the evolving mobility sector.

Industrial Internships

Industrial Internship – MEIL (Megha Engineering & Infrastructures Ltd.)

Duration: [10 March 2025 – 09 Apr 2025]

Role: Mechanical Engineering Intern

Key Responsibilities & Learnings:

- Worked on conventional manufacturing machines including lathe, milling, drilling, and CNC machines
- Participated in quality control and inspection using vernier calipers, micrometers, bore gauges, and dial indicators
- Assisted in interpreting engineering drawings and manufacturing tolerances
- Gained exposure to fabrication, machining processes, and industrial safety standards
- Collaborated with engineers during manufacturing and inspection stages

	Megha Engineering & Infrastructures Ltd. An ISO 9001-2015 Company S-2, Technocrat Indl. Estate, Balanagar, Hyderabad-500037, Telangana, INDIA Tel: +91-40-44336700 Fax: +91-40-44336800 E-mail: info@meil.in Visit us: www.meil.in U45202TG2006PLC050271
	Date: 16/04/2025 <u>TO WHOM SOEVER ITMAY CONCERN</u> This is to certify that Mr.Guduru Sudeep pursuing B.Tech-Mech(2021-25), Roll No.21QP1A0304, and he has undergone Internship on "Manufacturing" from 10/03/2025 to 09/04/2025 in our organization in the field of Mechanical. During the period of the internship in our organization, his conduct was found to be Good. We wish him all the best for his future endeavor. For Megha Engineering and Infrastructures Limited  M.Muralidhar Sr.Manager HR 
	

Industry Experience – KIA Motors (Service Division)

Organization: KIA Motors

Role: Service / Workshop Trainee

Duration: [June 2023 – July 2023]

Responsibilities & Exposure:

- Assisted in vehicle inspection and servicing procedures for vehicle engines.
- Gained practical understanding of engine systems, transmission, suspension, braking systems, and steering.
- Observed and supported diagnostic procedures using service tools and scan equipment.
- Participated in preventive maintenance activities, including fluid checks and component replacements.
- Learned real-world failure modes and customer-reported vehicle issues.

CATIA V5 Design Projects

Introduction

This section presents selected mechanical and automotive-oriented design projects developed using CATIA V5. The models demonstrate my understanding of part design, assembly constraints, kinematics, and manufacturability, with emphasis on automotive components and mechanical systems.

Project 1

Crankshaft and Piston Assembly

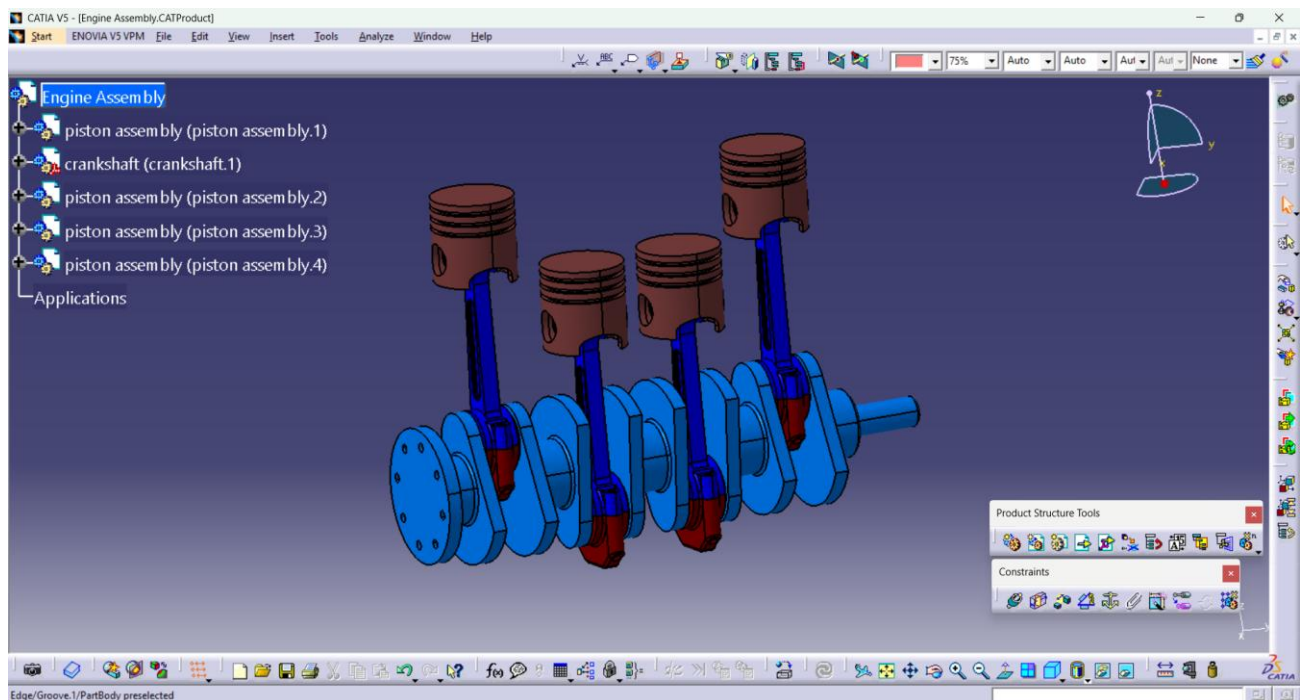
Software: CATIA V5 R19

Modules Used: Part design, Assembly Design

Application: Internal Combustion Engine Mechanism

Project Description

- Designed crankshaft, pistons, connecting rods, and rings
- Applied correct geometric constraints for smooth kinematic motion
- Maintained functional clearances between moving components
- Focused on manufacturability and assembly sequence



Project 2

Disc Brake Design

Software: CATIA V5 R19

Modules Used: Part design, Assembly Design

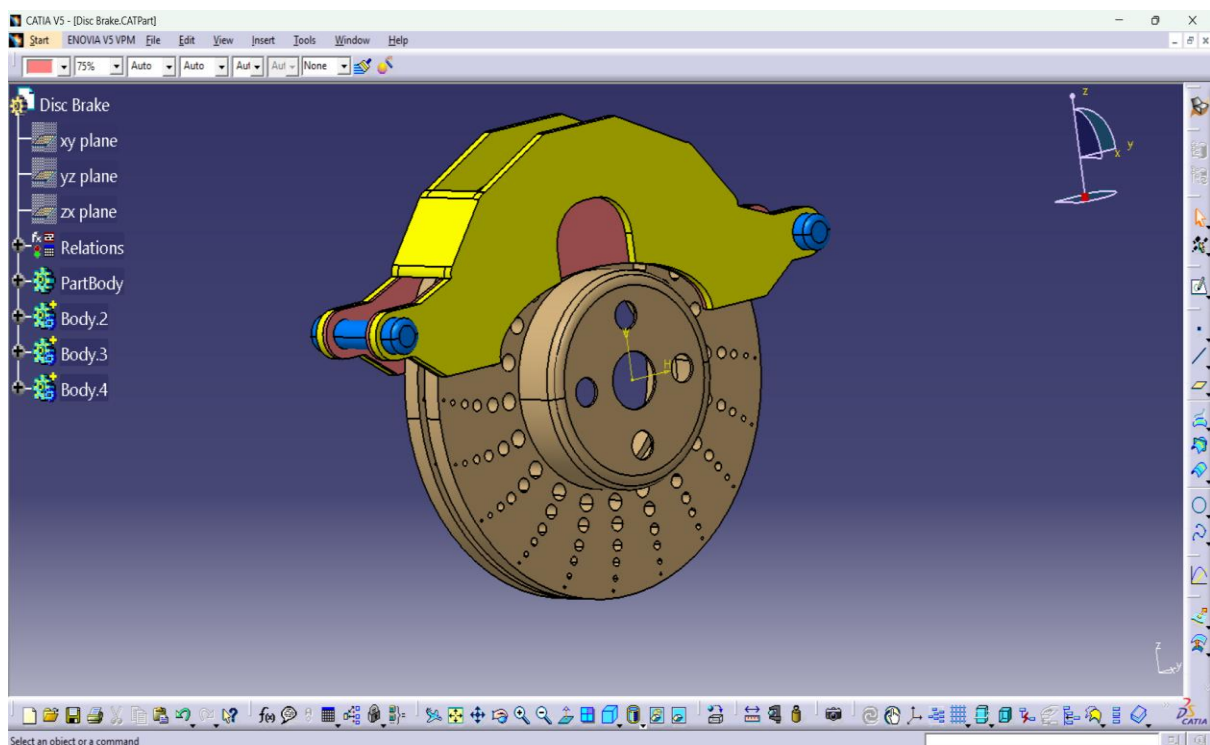
Application: Automotive Braking System

Project Description

- Designed a complete disc brake assembly, including brake disc (rotor), caliper, brake pads, hub, and mounting bracket
- Followed standard automotive braking system layout
- Maintained appropriate clearances between rotating and stationary components
- Assembly constraints applied to simulate correct mounting and alignment

Design Considerations

- Disc diameter and thickness based on braking requirement
- Caliper positioning for uniform pad pressure
- Manufacturability and ease of assembly



Final Year Major Project

Design and Fabrication of an Electric Vehicle

Degree: B.Tech – Mechanical Engineering

Type: Academic Major Project

Tools Used: CATIA V5, ANSYS, AutoCAD

Domain: Electric Vehicle / Automotive Engineering

Project Overview

- Designed and fabricated a prototype electric vehicle (EV) with emphasis on lightweight structure, efficiency, and practical feasibility
- The project involved concept design, CAD modeling, component selection, structural analysis, fabrication, and testing
- Focused on promoting sustainable and affordable electric mobility

Vehicle Design

- Complete 3D CAD model developed using CATIA V5

Designed major systems:

- Chassis / frame
- Motor mounting structure
- Battery placement

Design focused on:

- Structural safety
- Proper weight distribution
- Ease of fabrication and maintenance

Key EV Components

Motor: BLDC motor

Battery: Lithium-ion battery pack

Controller: Matching motor controller

Transmission: chain drive

Braking System: Mechanical braking system

Selection Criteria

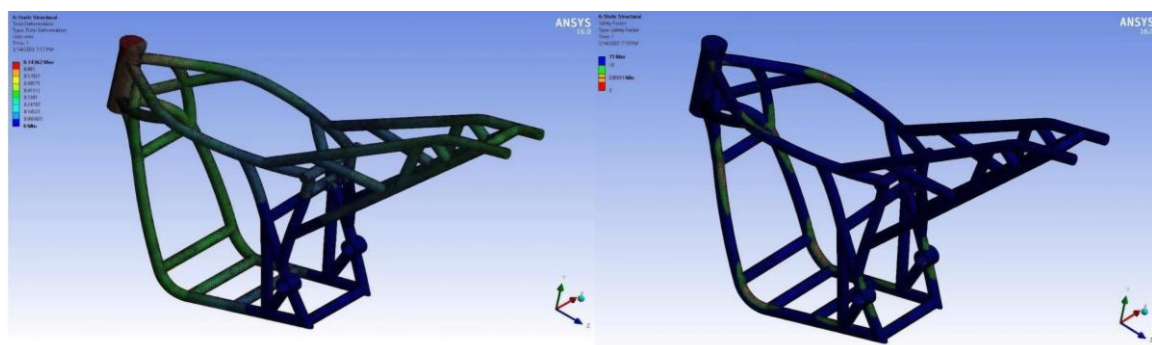
- Vehicle load & speed requirement
- Energy efficiency
- Cost and availability
- Safety considerations

Structural Analysis

- Conducted Finite Element Analysis (FEA) on chassis/frame using ANSYS
- Applied loading conditions:
 - Rider weight
 - Vehicle self-weight
 - Road assumptions

Verified:

- Stress distribution
- Deformation
- Factor of safety



- Deformation of chassis (mm)

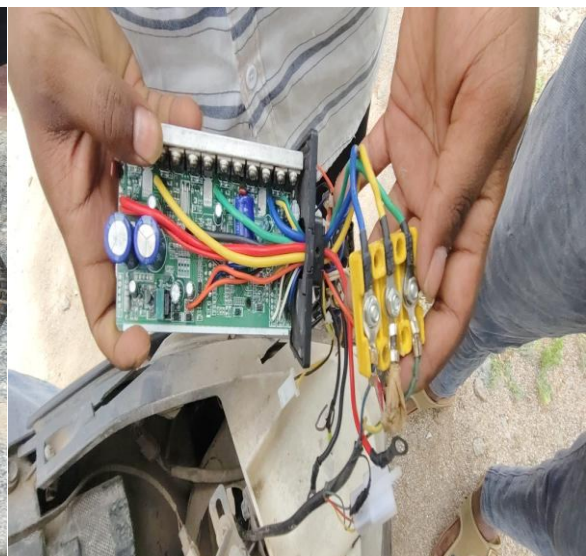
- A factor of safety of chassis

Fabrication Process

- Frame fabrication using welding techniques
- Mounting of motor, battery, controller, and braking system
- Electrical wiring and assembly

Tools Used

- Welding machine
- Hand tools
- Measuring instruments



Performance Results

Vehicle weight: 30kgs

Top speed: 60 kmph

Range per charge: 80kmph

Braking performance: tested under controlled conditions

Observations

Stable vehicle operation

Satisfactory braking and handling

Efficient energy usage for short-distance travel

Conclusion

- Successfully designed and fabricated a functional EV prototype
- Validated design through analysis and testing
- Demonstrated practical application of mechanical and automotive engineering principles

Key Learnings

- Electric powertrain fundamentals
- Vehicle system integration
- Design-for-manufacturing
- Importance of safety and reliability in EVs

Contact Information

Name: Guduru Sudeep

Degree: B.Tech – Mechanical Engineering (Final Year)

Specialization Interest: Automotive Engineering / Manufacturing/Design

Phone: +91 7816057828

Email: sudeepchowdary27@gmail.com

Location: Telangana, India

Technical Skills:

AutoCAD, CATIA V5, SolidWorks, Engineering Drawings

Experience:

Internship – KIA Service Center

Internship – MEIL (Manufacturing, QC, Design)